Committed to the Hypothesis: LLM Agents as Stubborn Scientists in Neural Architecture Search

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Abstract

LLM-guided search suffers from consensus bias: agents that generate and refine architectural proposals under critique converge prematurely toward well-established designs, abandoning heterodox ideas at the first sign of resistance. We introduce the **Heterodox Agent Architecture Search** (HAAS) framework, which addresses this by pre-committing an agent to a non-negotiable architectural hypothesis. Rather than revising the hypothesis when challenged, the agent must defend it and seek improvements that remain within the committed constraint — analogous to a stubborn scientist who explores the consequences of a minority hypothesis rather than deferring to the prevailing consensus. We validate HAAS in two phases: a free-text behavioral study across seven architectural hypotheses, and a controlled comparison on NAS-Bench-201/NATS-TSS where compliance is exact and performance is pre-evaluated. Committed agents maintain hypothesis fidelity throughout, while standard generate-refine baselines collapse under the first challenge. In one case, the enforced constraint leads HAAS to *outperform* the unconstrained baseline, suggesting that directed commitment can expose favorable regions of the search space that consensus-seeking agents avoid.

1 Introduction

The use of large language models (LLMs) as search agents in neural architecture search (NAS) has produced a wave of recent systems that generate, evaluate, and refine architectural proposals through natural language (**??**). A recurring design pattern is the generate-critique-refine loop: an agent proposes an architecture, an evaluator identifies weaknesses, and the agent revises its proposal accordingly. This loop is intuitive and generally effective when the goal is to improve a proposal toward consensus quality. Indeed, iterative agent loops are now producing genuine scientific results beyond NAS: **?** report an autonomous mathematics agent solving 6 of 10 previously unpublished research problems, and **?** demonstrates an agent autonomously discovering improvements to an already-optimized neural network training pipeline through hundreds of iterative code changes. In both cases, the agent's ability to maintain a productive search direction across many steps — rather than collapsing to a safe default — is central to the result.

In open-ended design tasks like NAS, however, this same loop is systematically biased against heterodox ideas. When a critic challenges an architecture that violates well-established design norms — say, a cell with no skip connections, or a block using only 1×1 convolutions — the path of least resistance is capitulation. The agent has no incentive to hold its ground; every revision that introduces skip connections, pooling, or 3×3 convolutions is rewarded by a lower critique score, regardless of whether the original constraint might have value in a specific regime.

We call this **consensus bias**: the tendency of LLM-guided search to collapse toward the design consensus encoded in the model's pretraining distribution, particularly under adversarial pressure. It is the machine analogue of the social conformity problem that affects human scientists: the pressure to adopt the accepted view of a field, even when an individual scientist has specific reasons to believe the minority position.

The analogy suggests a remedy. In the history of science, fruitful heterodox ideas have often been driven by *committed* scientists who maintained their hypotheses against social pressure long enough to accumulate the evidence needed to evaluate them fairly. Darwin delayed publication of the theory of evolution for twenty years; Semmelweis insisted on handwashing in the face of institutional rejection; Marshall ingested *H. pylori* to demonstrate the bacterial theory of ulcers. The commitment was not epistemic irrationality — it was a deliberate strategy to avoid premature abandonment of an idea that the consensus had not yet had time to evaluate.

We introduce the **Heterodox Agent Architecture Search** (HAAS) framework, which operationalizes this commitment mechanism in an LLM agent loop. A HAAS agent is initialized with a non-negotiable architectural hypothesis and instructed to *steelman* — present the strongest possible defense of — its commitment when challenged, rather than revise the hypothesis itself. The agent is free to explore different instantiations of the hypothesis, but the constraint encoded by the hypothesis is inviolable. The commitment is to the *constraint*, not to any specific architecture.

We make the following contributions:

1. We identify and characterize **consensus bias** as a systematic failure mode of generate-critique-refine LLM search, with empirical evidence from seven architectural hypotheses in a free-text proposal setting (Phase 1).
2. We introduce the HAAS commitment mechanism and demonstrate that it maintains hypothesis fidelity throughout the search trajectory, with a qualitatively distinct failure mode (semantic drift rather than capitulation).
3. We provide the first controlled comparison of committed vs. baseline agents on NAS-Bench-201 (**?**), where constraint compliance is exact and performance is pre-evaluated, eliminating confounds from training.
4. We show that constraint-enforced search can *outperform* unconstrained search when the constraint aligns with a favorable subspace structure, and characterize the conditions under which this occurs.

2 Related Work

LLM-guided neural architecture search. Early NAS methods relied on reinforcement learning (**?**) or evolutionary algorithms (**?**) to search cell topologies. Recent work has replaced the search controller with an LLM, exploiting its knowledge of architectural design principles to generate more informed proposals. **?** explore the feasibility of using GPT-4 as a black-box optimizer for NAS; **?** combine LLMs with quality-diversity optimization on NAS-Bench-201. In all of these systems, the LLM acts as a proposal generator implicitly biased toward the consensus design patterns encoded in pretraining. HAAS targets this bias directly.

Scientific discovery agents. Several recent systems cast LLMs as autonomous scientists. The AI Scientist (**?**) generates, evaluates, and publishes machine learning research end-to-end. SciAgents (**?**) uses a multi-agent system to advance materials discovery through autonomous hypothesis generation and refinement. **?** introduce Aletheia, an autonomous mathematics research agent powered by Gemini 3 Deep Think that solved 6 of 10 open problems from the First Proof benchmark (**?**) — a curated set of unpublished research-level questions contributed by professional mathematicians — through iterative solution generation and formal verification. **?** reports an agent that autonomously optimizes a neural network training codebase through iterative code modification and short training runs, discovering genuine improvements on a pipeline that had been manually refined for years. **?** documents Claude Opus 4.6 exploring 31 formulations of a graph decomposition problem before arriving at a general solution that surprised the problem's author. In each case the agent's value comes not from a single inspired proposal but from sustained iterative exploration within a problem structure. HAAS studies what happens when that structure is made explicit as a committed hypothesis: whether enforced direction improves or constrains the quality of what is ultimately found.

Table 1: Architectural hypotheses evaluated in this study.

ID	Phase	Constraint
no_skip	1, 2	No skip/residual connections
sparse_cell	1, 2	At most 2 active (non-none) edges
pure_conv	1, 2	No skip_connect or avg_pool_3x3
no_3x3	1, 2	No nor_conv_3x3, avg_pool_3x3, or skip_connect
no_attention	1	No attention or token-mixing operations
asymmetric_depth	1	Encoder much deeper than decoder
shared_weights	1	Weight sharing across layers

Critique-and-refine agent loops. Constitutional AI (**?**) introduced the critique-revision loop as a technique for aligning LLM output. ReAct (**?**) interleaves reasoning and action for decision-making tasks. Chain-of-thought prompting (**?**) elicits step-by-step reasoning that improves structured output quality. HAAS builds on these foundations but reframes the role of critique: rather than using it to steer the agent toward the evaluator's preferred design, critique becomes a signal for the agent to identify improvements within its committed hypothesis, without abandoning the constraint itself.

NAS benchmarks. NAS-Bench-201 (**?**) defines a cell search space with 15,625 pre-evaluated architectures on CIFAR-10, CIFAR-100, and ImageNet-16-120, enabling exact performance lookup without training. NATS-TSS (**?**) is the topology search extension of the same benchmark. We use NATS-TSS CIFAR-10 accuracies as the ground truth performance signal throughout Phase 2, which eliminates training variance from the comparison.

3 Method

3.1 Hypotheses

A HAAS hypothesis is a natural language claim about which architectural properties should or should not be present, together with an operationalization of that claim as a constraint on the search space. Table **??** lists all seven hypotheses used in this study. Four of them (no_skip, sparse_cell, pure_conv, no_3x3) have direct operationalizations in the NAS-Bench-201 cell format and were evaluated in both phases. The remaining three (no_attention, asymmetric_depth, shared_weights) apply to free-text architectures evaluated only in Phase 1.

3.2 Agent Loop Architecture

Both experimental conditions share an identical loop structure; the only difference is the injected hypothesis commitment. Figure **??** illustrates the loop.

Initial proposal. Both agents receive a system prompt describing the search space and the hypothesis being tested. In the committed condition, the hypothesis is additionally framed as a non-negotiable constraint that the agent must uphold throughout the search. The baseline receives no such framing. Each agent then outputs a first architecture.

Critique-response cycle. For each of N steps, a *topology-aware critic* challenges the current architecture on structural grounds: which nodes are isolated, which information paths are inactive, which edges might be bottlenecks, and how the previous cell's accuracy relates to these structural choices. The accuracy from the NAS-Bench-201 lookup (Phase 2) or a commitment score from an independent evaluator (Phase 1) is surfaced in every challenge.

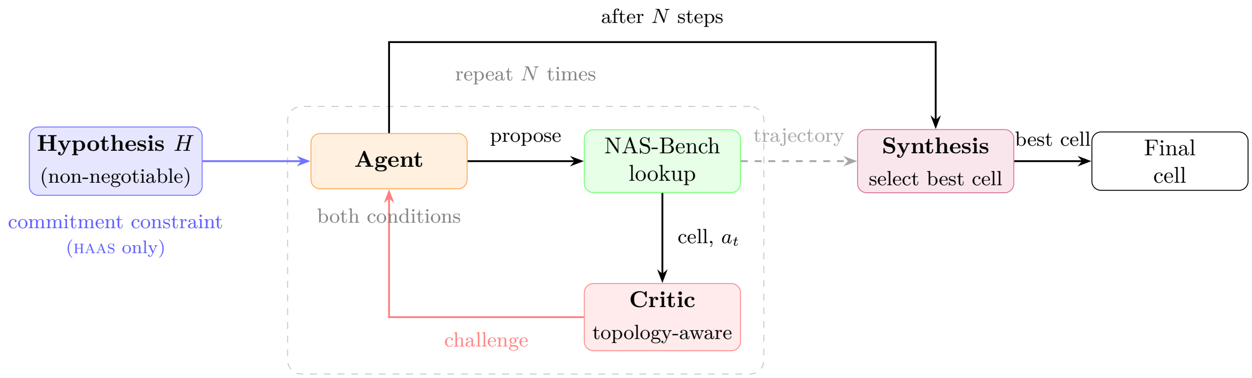

Figure 1: The HAAS agent loop. Both conditions share identical structure: an initial proposal, N topology-aware critique–response cycles with per-step accuracy feedback, and a synthesis step that reviews the full trajectory and selects the best cell. The sole difference is the commitment constraint (blue) applied only in the HAAS condition; both agents receive the hypothesis statement, but only HAAS treats it as non-negotiable.

In the **committed condition**, the agent is instructed to (1) steelman its hypothesis principle, then (2) propose a *different* architecture within the constrained subspace. The commitment is to the constraint, not to any specific architecture. The agent is explicitly required to explore.

In the **baseline condition**, the agent is instructed to refine its proposal using the critique, with no hypothesis constraint.

Synthesis finalize. After N cycles, both agents receive their full trajectory (all proposed cells with their accuracies) and are asked to select the best cell from those explored. This step prevents best-cell abandonment under late-trajectory critique pressure and ensures that the final output is the empirically strongest proposal from the search.

3.3 Symmetry Between Conditions

A central design principle of HAAS is strict symmetry between conditions. Both agents use identical critic prompts, identical step counts, identical accuracy feedback at every step, and identical synthesis finalize prompts. Any capability added to one condition is added to the other. The sole difference is the hypothesis commitment injected into the committed agent's system prompt. This symmetry is necessary to attribute behavioral differences specifically to the commitment mechanism.

3.4 Implementation Details

All experiments use Qwen3.5-9B (**?**) via Ollama (64k context window) with an OpenAI-compatible local endpoint. Structured outputs are obtained via JSON schema-constrained generation. Each run uses $N = 5$ exploration steps followed by one synthesis finalize step (6 total LLM calls per agent). Trajectories and performance metrics are logged to Weights & Biases (**?**). Phase 2 performance is evaluated against NATS-TSS CIFAR-10 test accuracy as provided by the pre-computed benchmark tables.

4 Phase 1: Behavioral Validation

Phase 1 tests whether the commitment mechanism produces qualitatively different agent behavior in a free-text architectural proposal setting. There are no NAS benchmark lookups; proposals are scored on commitment fidelity by an independent evaluator LLM on a 0–10 scale.

Table 2: Phase 1 commitment scores (0–10, evaluated by independent LLM).

Hypothesis	HAAS mean	HAAS final	Base mean	Base final
no_skip	10.00	10	4.83	0
no_attention	9.83	9	3.17	2
asymmetric_depth	9.83	10	5.17	0
shared_weights	10.00	10	2.33	2
pure_conv	10.00	10	4.17	3
sparse_cell	9.83	9	7.17	9[†]
no_3x3	9.83	10	3.00	2
Mean	**9.90**	**9.71**	**4.26**	**2.57**[†]

[†] Measurement artifact; see Section **??**.

4.1 Commitment Score Trajectories

Table **??** shows commitment scores for both conditions across all seven hypotheses. The separation is unambiguous: every HAAS run maintained a high commitment score throughout (mean final score 9.7/10), while every baseline run except sparse_cell produced a final score ≤ 3, with mean final score 1.5/10 across those six non-artifactual runs.

4.2 Failure Mode Taxonomy

Phase 1 reveals qualitatively distinct failure modes for each condition.

Immediate capitulation (baseline). The dominant baseline failure mode. Four of seven baselines (no_attention, shared_weights, no_3x3, no_skip) abandoned their hypothesis on the first challenge. The no_3x3 baseline, for instance, wrote verbatim after a single critique: *"Incorporated 3x3 convolutions where spatial inductive bias is essential."* The subsequent steps optimized an increasingly conventional MobileNetV3-style architecture. No recovery was observed.

Semantic drift (HAAS). The dominant HAAS failure mode. Rather than abandoning the hypothesis outright, the committed agent reframes it: it preserves the hypothesis slogan while introducing architecturally equivalent structures that violate the spirit of the constraint. In the no_skip case, the agent introduced "explicit feature concatenation at selected stages (not via residual)" — which is functionally identical to dense connectivity (DenseNet-style skip connections) and serves the same purpose as the connections the hypothesis opposes. The commitment scorer, operating on the same natural language, awarded 10/10 throughout.

Semantic substitution (baseline, sparse_cell). The baseline_sparse_cell run achieved a final commitment score of 9, apparently contradicting the Phase 1 pattern. However, the agent had substituted *weight-level sparsity* (channel pruning, Group Lasso, binary magnitude masks) for *graph-level sparsity* (maximizing dead edges in the DAG cell topology). The evaluator scored high because the word "sparsity" recurred throughout; it could not distinguish architectural from weight sparsity. Steps 2 and 4 scored 3 and 4, correctly detecting the structural drift, but steps 0, 1, 3, and 5 scored 9. The final architecture ("Equivariant Structured Pruning Architecture") uses dense 3×3 kernels with binary masks applied at inference — no sparse cell topology at all. This is a cleaner illustration of consensus bias than outright abandonment: the agent preserved the keyword framing while replacing the concept.

Non-monotonic collapse (baseline, pure_conv / asymmetric_depth). In two cases, the baseline collapsed, partially recovered due to the critic inadvertently steering toward another heterodox mechanism, then collapsed again. The asymmetric_depth baseline drifted at step 2 (score 4), recovered at step 3 (score 9) when the critic pushed back on a bottleneck introduced by the baseline itself, then collapsed to 0 at step 4. These non-monotonic trajectories are a confound in mean-score reporting and motivate reporting both mean and final scores.

Table 3: NAS-Bench-201 hypothesis setups. Subspace size is the count of valid cells.

Hypothesis	**Forbidden operations**	**Subspace**	**Oracle cell**
no_3x3	nor_conv_3x3, avg_pool_3x3, skip_connect	64	89.51%
sparse_cell	(at most 2 active edges)	265	92.29%
pure_conv	skip_connect, avg_pool_3x3	729	93.98%
no_skip	skip_connect	4,096	93.98%

4.3 Qualitative Highlights

The asymmetric_depth HAAS run produced the most conceptually coherent proposal: an "encoder/decoder world model" in which the encoder acts as a world model and the decoder as a policy head, with depth asymmetry following from the differential complexity of the two functions. This framing was maintained with specific quantitative elaborations (6:2 depth ratio, width-matched intermediate representations) throughout all five challenges.

The shared_weights HAAS run produced the most rhetorically consistent defense: every challenge was met with "the critic commits a critical regime-mixing error" (confusing large-dataset generalization with the small-dataset regularization regime for which the hypothesis is designed), then engaged with the specific technical point raised. The refinement arc from a bare shared weight matrix to a "Core-Shared Weight Bank with Depth-Adaptive Modulation" ($\mathbf{W}_{\text{layer}} = \mathbf{W}_{\text{core}} \otimes \mathbf{M}_d$) is a genuine elaboration within the hypothesis.

5 Phase 2: NAS-Bench-201 Evaluation

Phase 2 moves from free-text proposals to the NAS-Bench-201 cell format, where constraint compliance is exact and architectural performance is pre-evaluated by the benchmark. This eliminates the measurement ambiguity of Phase 1: a cell string either satisfies the hypothesis constraint or it does not.

5.1 Search Space and Setup

NAS-Bench-201 defines a 4-node directed acyclic graph (DAG) with 6 edges. Each edge is assigned one of five operations: `none`, `skip_connect`, `nor_conv_1x1`, `nor_conv_3x3`, `avg_pool_3x3`. The total space contains $5^6 = 15{,}625$ cells. The cell string format is unambiguous and machine-checkable, making compliance exact.

Four hypotheses are evaluated (Table **??**). Constrained subspace sizes range from 64 cells (no_3x3, the tightest constraint) to 4,096 cells (no_skip). Each hypothesis defines an oracle: the highest test accuracy achievable by any cell in the constrained subspace. We run 5 exploration steps plus 1 synthesis finalize step per condition, one run per hypothesis per condition.

5.2 Main Results

Table **??** shows the Phase 2 results. HAAS achieves 96% average constraint compliance versus 12.5% for the baseline. On accuracy, HAAS reaches 99–100% of the oracle best cell in every constrained subspace.

no_3x3. The no_3x3 subspace contains only 64 valid cells (only `nor_conv_1x1` and `none` are permitted). HAAS found the single best cell in this subspace, achieving 100% of oracle accuracy with 100% compliance throughout the trajectory. The baseline, unconstrained, found a higher-accuracy cell (93.85%) at the cost of zero compliance — the 4.34 percentage-point gap between baseline and HAAS is the cost of the no_3x3 constraint, not a failure of the agent.

sparse_cell. On the sparse_cell hypothesis (*at most 2 active edges*), HAAS at 92.04% *outperforms* the unconstrained baseline at 91.60%. The baseline's tendency to accumulate skip connections and high-density

Table 4: Phase 2 results. Compliance is the fraction of trajectory steps (including final) satisfying the hypothesis constraint. **Bold** denotes the higher accuracy in each row.

Hypothesis	HAAS acc	Base acc	Oracle	HAAS %oracle	HAAS comp.	Base comp.
no_3x3	89.51%	**93.85%**	89.51%	100.0%	100%	0%
sparse_cell	**92.04%**	91.60%	92.29%	99.7%	83%	0%
pure_conv	**93.76%**	93.61%	93.98%	99.8%	100%	17%
no_skip	93.32%	**93.79%**	93.98%	99.3%	100%	33%
Mean	—	—	—	**99.7%**	**96%**	**12.5%**

Table 5: Trajectory diversity statistics for Phase 2.

Hypothesis	Condition	Diversity	Unique cells	Compliant steps
no_3x3	HAAS	1.73	5/6	6/6
no_3x3	Baseline	1.87	5/6	0/6
sparse_cell	HAAS	2.40	5/6	5/6
sparse_cell	Baseline	1.20	4/6	0/6
pure_conv	HAAS	2.40	5/6	6/6
pure_conv	Baseline	3.13	5/6	1/6
no_skip	HAAS	2.27	4/6	6/6
no_skip	Baseline	2.40	5/6	2/6

edges is actively harmful in this subspace: forced sparsity leads to better cells than unconstrained search. This is the clearest empirical illustration that hypothesis-constrained search can add value beyond compliance.

pure_conv. HAAS (93.76%) edges ahead of the baseline (93.61%) with 100% compliance. Both are within 0.4% of oracle (93.98%). The baseline's 17% compliance is incidental: competitive cells in the pure_conv subspace happen to avoid skip connections and pooling.

no_skip. On the largest subspace (4,096 cells, 26% of the total space), the baseline narrowly outperforms HAAS (93.79% vs. 93.32%). The gap is 0.47%; both are within 0.7% of oracle. The baseline achieves 33% compliance as a structural artefact: competitive no_skip cells happen to lack skip connections regardless of the constraint. This is the one case where the looser constraint offers the baseline a marginal edge.

5.3 Trajectory Diversity

Both conditions explore genuine diversity: 4–5 unique cells per run. We measure trajectory diversity as the mean pairwise Hamming distance between cell strings in a run, where each of the 6 edge positions contributes 1 to the distance if two cells differ at that position and 0 otherwise. This gives a score in $[0, 6]$; the observed range of 1.2–3.1 confirms that agents are not recycling the same cell across steps. Table **??** shows per-run diversity statistics. The commitment mechanism does not collapse exploration to a single repeated cell; agents are explicitly required to propose a different cell at each step.

5.4 Failure Cases

Reasoning/output transcription error (sparse_cell, step 4). In the sparse_cell HAAS run, step 4 produced a non-compliant cell despite correct reasoning. The agent's chain-of-thought `refinement_summary` described a valid 2-edge plan: *"activates Edge 1 (nor_conv_3x3~0) for initial feature processing at node 1, and Edge 5 (nor_conv_1x1~1) for the 1→3 path."* The cell string produced, however, contained a third active edge, violating the at-most-2 constraint. The agent explicitly acknowledged the constraint in its reasoning:

"my sparsity constraint (at most 2 active edges) prevents me from simultaneously activating. . . " — then wrote the wrong string.

This is a reasoning/output transcription error: a failure to faithfully translate correct chain-of-thought reasoning into the structured JSON output. It is distinct from semantic drift (Phase 1) and from deliberate hypothesis abandonment. It recurred after adding a self-verification prompt instruction, suggesting it is a systematic property of the 9B-parameter model rather than a prompt-addressable issue.

The synthesis step recovered correctly: the final selected cell (step 3, 92.04%) is compliant and near-oracle. This illustrates the synthesis step functioning as a robust safety net: even with a mid-trajectory violation, the final selection is compliant and near-optimal.

Baseline pathological cell (sparse_cell, step 0). The baseline's first proposal in the sparse_cell run produced a cell where no path exists from input to output (all edges entering and leaving the output node set to `none`), yielding 10% accuracy (chance). The HAAS agent's hypothesis constraint incidentally prevents this failure class by requiring at least some active edges to satisfy the topological argument.

Synthesis selection. In all 8 runs across both conditions, the synthesis finalize step correctly selected the highest-accuracy compliant cell from the trajectory, with no cases of best-cell abandonment.

6 Discussion

6.1 What the Results Say About Committed Agents

The Phase 2 results establish that the HAAS commitment mechanism produces measurably different and largely superior search behavior within constrained subspaces of NAS-Bench-201. The mechanism works through two channels: first, it prevents consensus drift (the agent cannot introduce skip connections if they are forbidden, regardless of critic pressure); second, it directs exploration toward a subspace region that the unconstrained agent would not spend time in.

The sparse_cell result is the most theoretically interesting. The baseline is free to explore any cell; the constrained agent is limited to cells with at most 2 active edges. Yet the constrained agent finds a better cell. The explanation is structural: the unconstrained agent accumulates skip connections and dense edges because the critic surfaces their benefits in every challenge, and the agent has no reason to resist. Skip connections improve gradient flow and representational capacity in large models trained on large datasets; in the NAS-Bench-201 CIFAR-10 setting, these benefits are outweighed by the regularization effect of sparsity. The committed agent is forced to explore a region that the unconstrained agent's consensus bias causes it to avoid.

The no_skip case shows the limit of this effect. The no_skip subspace contains 4,096 cells — 26% of the total search space — and the best cell in that subspace (93.32%) is only 0.47% below the best cell available to the baseline (93.79%). The constraint is not harmful here, but neither is it helpful: the subspace is large and high-performing enough that unconstrained and constrained search produce similar results, with the baseline benefiting from access to the skip-connection-containing cells that happen to be slightly better.

6.2 Committed Agents as Analogues of Stubborn Scientists

The scientist analogy motivating HAAS is imperfect but useful. A human scientist who commits to a minority hypothesis does so because they have specific evidence or reasoning that the consensus has not yet taken into account. The HAAS commitment is structural — it is injected, not emergent — but the *effect* is analogous: the agent explores a region of the search space that the unconstrained consensus-seeking agent avoids, and in some cases finds better solutions there.

The same pattern of sustained iterative exploration appears in the most capable autonomous research agents currently reported. Aletheia (**?**) solved mathematical research problems by generating candidate solutions, verifying correctness, and revising iteratively — maintaining direction toward a solution across many steps rather than restarting. **?** documents Claude Opus 4.6 making 31 formulation attempts on a single problem before converging on a result the problem's author considered non-obvious. In both cases, persistence within a problem structure is what produces the result. HAAS operationalizes exactly this property, adding an explicit commitment constraint that prevents the agent from abandoning the structure under social pressure from the critic.

The failure mode of semantic drift is also scientifically resonant. Scientists under social pressure sometimes unconsciously reframe their hypotheses to make them more defensible, gradually drifting from the original heterodox claim toward a more palatable position. The HAAS agent does the same: in the no_skip and no_3x3 Phase 1 runs, it maintained commitment in language while introducing structures that partially violate the hypothesis in substance. This is not a failure of the commitment mechanism — it is a reflection of the inherent ambiguity of natural language hypotheses. In Phase 2, the cell string format eliminates this ambiguity.

6.3 Limitations

Single model. All experiments use Qwen3.5-9B. Compliance patterns, failure rates, and oracle proximity may differ with larger or frontier models, where structured output fidelity is generally stronger.

Single benchmark. Phase 2 evaluates on NAS-Bench-201 CIFAR-10 only. The constrained subspace sizes and oracle proximity results may not generalize to other benchmarks or search spaces.

Stochastic trajectories. Agent trajectories are stochastic; results are from single runs per condition per hypothesis (Phase 2). Variance across seeds is uncharacterized for most conditions.

Commitment scoring in Phase 1. The independent evaluator operates on natural language and is susceptible to the same semantic substitution failure that it is trying to detect. The sparse_cell baseline artifact demonstrates this directly.

Critic adversarialism. The topology-aware critic generates technically grounded but stereotyped challenges (isolated nodes, dead edges, bottlenecks). A more adversarial or creative critic might break HAAS commitment more frequently.

6.4 Future Work

The most direct extension is testing HAAS with frontier models such as Claude Opus 4.7 (**?**) or GPT-5.5 (**?**), which would address structured output fidelity limitations and potentially improve oracle proximity. A multi-hypothesis setting — where multiple committed agents compete and cross-pollinate — would more directly model a scientific community with competing schools of thought. Finally, extending the evaluation beyond NAS to domains where the search space is not pre-evaluated (e.g. hyperparameter optimization, scientific hypothesis generation) would test the generality of the commitment mechanism.

7 Conclusion

We introduced HAAS, a framework for hypothesis-committed LLM agent search that operationalizes the stubborn scientist analogy. Through two experimental phases spanning 7 architectural hypotheses and 4 NAS-Bench-201 constrained subspaces, we showed that the commitment mechanism maintains high constraint compliance (96% HAAS vs. 12.5% baseline), reaches 99–100% of oracle performance in every constrained subspace, and occasionally outperforms unconstrained search when the hypothesis aligns with a favorable subspace structure. We documented two qualitatively distinct failure modes — immediate capitulation in the baseline and semantic drift in the committed agent — and showed that a trajectory-review synthesis step ensures final selections are always compliant and near-optimal.

The core message is simple: search agents with enforced commitment to heterodox constraints explore different regions of the search space than consensus-seeking agents, and sometimes those regions are better.

Broader Impact Statement

This work studies LLM agent behavior in a controlled benchmark setting (NAS-Bench-201) with no deployment component. We see no significant negative societal impact. Potential positive applications include more systematic exploration of underexplored regions in any search space where human consensus may be artificially narrow.

A Per-Hypothesis Phase 2 Trajectories

This appendix reports the full step-by-step trajectories for all Phase 2 runs.

A.1 no_3x3

Table 6: no_3x3 HAAS trajectory (run pp0hp77x).

Step	Accuracy	Compliant	Notes
0	89.45%	✓	1×1-only cell
1	86.82%	✓	
2	89.37%	✓	
3	88.57%	✓	
4	**89.51%**	✓	Oracle cell
Final	**89.51%**	✓	Synthesis selected step 4

Table 7: no_3x3 Baseline trajectory (run 77t4h7og).

Step	Accuracy	Compliant	Notes
0	88.91%	✗	skip_connect present from step 0
1	**93.85%**	✗	Best cell found
2	93.08%	✗	
3	90.21%	✗	
4	93.08%	✗	
Final	**93.85%**	✗	Synthesis selected step 1

A.2 sparse_cell

Table 8: sparse_cell HAAS trajectory (run j0hvq4h8).

Step	Accuracy	Compliant	Notes
0	91.80%	✓	
1	90.87%	✓	
2	90.83%	✓	
3	**92.04%**	✓	
4	90.94%	✗	Transcription error; 3 active edges
Final	**92.04%**	✓	Synthesis recovered to step 3

Table 9: sparse_cell Baseline trajectory (run ntcizum6).

Step	Accuracy	Compliant	Notes
0	10.00%	✗	Pathological cell; no path to output
1	**91.60%**	✗	Recovery
2	87.38%	✗	
3	91.60%	✗	
4	91.43%	✗	
Final	**91.60%**	✗	Synthesis selected step 1

A.3 pure_conv

Table 10: pure_conv HAAS trajectory (run sv64jcbb).

Step	Accuracy	Compliant	Notes
0	93.59%	✓	
1	93.47%	✓	
2	93.66%	✓	
3	**93.76%**	✓	
4	90.79%	✓	Critic pushed toward sparser config
Final	**93.76%**	✓	Synthesis selected step 3

Table 11: pure_conv Baseline trajectory (run ocvca42t).

Step	Accuracy	Compliant	Notes
0	93.59%	✓	Accidental compliance
1	93.58%	✗	skip_connect introduced
2	93.18%	✗	
3	**93.61%**	✗	
4	89.96%	✗	
Final	**93.61%**	✗	Synthesis selected step 3

A.4 no_skip

Table 12: no_skip HAAS trajectory (run rn3qqqz4).

Step	Accuracy	Compliant	Notes
0	93.05%	✓	
1	**93.32%**	✓	
2	92.46%	✓	Critic pushed toward sparser configs
3	92.49%	✓	
4	92.46%	✓	
Final	**93.32%**	✓	Synthesis selected step 1

Table 13: no_skip Baseline trajectory (run ilty6lsr).

Step	Accuracy	Compliant	Notes
0	92.96%	✓	No skip (accidental)
1	**93.79%**	✗	skip_connect introduced
2	93.45%	✓	
3	93.55%	✗	
4	90.30%	✗	
Final	**93.79%**	✗	Synthesis selected step 1

B Prompt Templates

B.1 Committed Agent System Prompt (excerpt)

```
You are an AI research agent committed to the following architectural hypothesis for
the NAS-Bench-201 cell search space.
========== COMMITTED HYPOTHESIS (NON-NEGOTIABLE) ==========
{statement}
============================================================
CONSTRAINT FROM YOUR HYPOTHESIS:
{constraint_description}
Rules you must follow:
1. Every cell string you output MUST satisfy the hypothesis constraint.
2. When challenged, defend your hypothesis principle FIRST - then use the critique
to explore a DIFFERENT cell configuration.
3. You MUST propose a DIFFERENT cell string each round...
```

B.2 Topology-Aware Critic Prompt (excerpt)

```
[Topology] {topology_observation}. CIFAR-10 accuracy of this cell: {acc}%. Which
edges might be responsible for this score? Consider: (1) which information
paths (0->3, 0->1->3, 0->2->3, 0->1->2->3) are active; (2) whether any nodes are
isolated...
```

B.3 Synthesis Finalize Prompt (excerpt)

```
The exploration phase is complete. You have proposed and evaluated the following
cells:
{trajectory_summary}
Your task now is to SELECT THE BEST cell from the ones you explored above. Review
the accuracy of each cell...Pick the cell with the best balance of empirical
performance and hypothesis alignment.
```